



Protecting intellectual property and ensuring foolproof chip applications

Semiconductor manufacturing operates within a high-stakes and fiercely competitive environment, where the integrity of intellectual property (IP) and the security of semiconductor chips are absolute priorities. As technology advances at an ever-accelerating pace, so too do the threats that loom ominously on the horizon, capable of compromising the confidentiality of susceptible chip designs and innovative breakthroughs. In this dynamic and rapidly evolving landscape, the protection of valuable intellectual assets and the assurance of chip security remain critical imperatives that demand constant vigilance and innovative defences.

Imagine the nightmare scenario of an attacker gaining access to a chip's design and production process, potentially stealing valuable IP and even counterfeiting products. Thankfully, there is a solution: split manufacturing. This innovative approach divides your chip's design into two distinct components, offering enhanced protection:

Front end. This component demands advanced manufacturing capabilities and is often outsourced to trusted foundries like TSMC and Global Foundries in Europe and Taiwan.

Back end. In contrast, the back end involves a more straightforward implementation that can be carried out locally, in your home country or facility.

Once both components are individually manufactured, they are integrated to create the final chip. This approach ensures that no single entity can access the complete design, significantly reducing the risk of IP theft and counterfeiting.

Hardware security: Strengthening chip defence

While software-based security measures, such as antivirus programs, undeniably serve as vital components of safeguarding digital environments, it is equally imperative to direct attention toward hardware-level security. Hardware-based security measures offer an additional and indispensable layer of defence against the diverse array of potential threats that exist in the ever-evolving digital landscape. In the following discussion, we will embark on a comprehensive exploration of these hardware-centric security strategies, elucidating their significance and efficacy in fortifying digital systems and data against malicious actors and vulnerabilities.

Removing JTAG and serial communication interfaces. Consider physically eliminating JTAG and serial communication interfaces from your products or boards if they are not required for reprogramming. These interfaces can be gateways for attackers to manipulate microcontroller or microprocessor binaries. This practice is widely accepted and helps prevent unauthorised code tampering.

Validating firmware authenticity with CRC checks. Implement cyclic redundancy checks (CRC) to verify the authenticity of firmware updates. CRC checks are straightforward operations that use adders and multipliers already present in your chip. During wireless transfers, CRC checks can detect accidental changes or tampering, ensuring only authorised firmware updates are applied.

Leveraging memory management units (MMU) and memory protection units (MPU). These hardware features are prevalent in modern processors and microcontrollers, offering enhanced security by isolating critical data from potential threats. MMUs and MPUs restrict a process's access to specific memory regions,

improving security.

Encryption for secure firmware distribution. When distributing firmware updates, encryption prevents unauthorised access and tampering. Many modern processors come equipped with encryption and decryption modules, simplifying the secure distribution of firmware updates.

Protecting intellectual property and chip applications is paramount, with split manufacturing emerging as a solution. Hardware-based security measures, such as encryption and memory management, complement software safeguards. Collaboration, research, and vigilance are vital for preserving chip integrity and security in our ever-evolving digital landscape.

In the competitive world of semiconductor manufacturing, safeguarding intellectual property is non-negotiable. Employing strategies such as split manufacturing and implementing hardware-level security measures ensures that your designs and innovations remain secure from malicious actors and counterfeiters. By adhering to these practices, you can fortify your chip security, protect your intellectual property, and maintain the trust of your customers and partners. In an era where technology is advancing at an unprecedented pace, staying one step ahead of potential threats is essential to maintain the integrity of your products and the security of your intellectual assets. In semiconductor manufacturing, securing electronic chips is crucial as they power our daily devices. However, hardware Trojans pose a persistent threat by stealthily infiltrating chip designs during fabrication. Strategies like enhanced testing, side-channel analysis, and security hardening can mitigate these threats, but challenges persist. **EFY**

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